

Finite response matrices for Weil positivity

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Abstract

We formulate an explicit matrix conjecture equivalent to the Riemann hypothesis. For each bounded interval, a digamma resolvent expansion and a Neumann comparison give an unconditional positive lower bound on a specified high-frequency sector of the Weil form. Eliminating that sector leaves a finite Hermitian response matrix, including every coupling to the low modes. We derive an explicit bounded, noncompact boundary correction to the diagonal Neumann model and convergent bounds for its action on finite input spaces. These yield rigorous lower and upper bounds on the response matrix and an equivalent formulation with arbitrarily small positive tolerances. The unresolved conjecture is positivity of these matrices on an unbounded sequence of intervals. The equivalence and the analytic reduction do not establish that positivity. No priority is claimed for the general finite-codimension or Schur-complement principles.

1 The proposed conjecture and its scope

The useful question from the longer investigation [6] is whether all possible negativity of the localized Weil form can be isolated in an explicit, finite response problem. The answer to the reduction question is yes. The arithmetic sign of the resulting matrices remains open.

The ingredients below define a matrix S_n for every integer $n \geq 1$ without using zeta zeros or assuming their location. Its dimension is a prescribed integer $N(n)$ obtained from an explicit high-mode estimate.

Conjecture 1.1 (Finite response). For every positive integer n , the matrix S_n defined in (4.1) is positive semidefinite.

In energy language, allowing the positive high modes to respond to a low-mode input never removes more than the low input's original energy. Theorem 4.1 proves that this assertion is equivalent to RH.

Localized Weil positivity, positivity on suitable subspaces of finite codimension, and variational formulations have substantial antecedents in Yoshida and Bombieri, as reviewed in [1, Section 1]. Explicit finite matrices also appear in [2]. The content to assess here is the particular Neumann boundary representation, the retained mixed response, and its controlled remainder. The Schur complement itself is standard. A comparison with those antecedents is necessary before making a novelty claim.

2 The arithmetic operator and an unconditional cutoff

Use $X_L = L^2(0, L)$, zero extension E_L , and $\widehat{F}(\tau) = \int_{\mathbb{R}} F(x) e^{-i\tau x} dx$. Inner products are antilinear in the first slot. Set

$$a_k = 2k + \frac{1}{2}, \quad b(s) = \sum_{k=0}^{\infty} \frac{2}{a_k} \frac{s}{a_k^2 + s} = \operatorname{Re} \psi\left(\frac{1}{4} + i\sqrt{s}/2\right) - \psi\left(\frac{1}{4}\right), \quad (2.1)$$

$$w_0 = \psi\left(\frac{1}{4}\right) - \log \pi, \quad \mathcal{D}_L = \left\{ f \in X_L : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{E_L f}(\tau)|^2 d\tau < \infty \right\}. \quad (2.2)$$

Let T_L be the nonnegative operator associated with the closed form

$$t_L[f] = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{E_L f}(\tau)|^2 d\tau.$$

For $d > 0$, $U_d F(x) = F(x - d)$ and $V_d = E_L^* U_d E_L$. Put $c_L(x) = \cosh((x - L/2)/2)$ and $s_L(x) = \sinh((x - L/2)/2)$. The bounded remainder and the full operator are

$$R_L = w_0 I + 2c_L c_L^* - 2s_L s_L^* - \sum_{\substack{p \text{ prime, } m \geq 1 \\ m \log p < L}} (\log p) p^{-m/2} (V_{m \log p} + V_{m \log p}^*), \quad (2.3)$$

$$W_L = T_L + R_L, \quad Q_L[f] = t_L[f] + \langle f, R_L f \rangle. \quad (2.4)$$

Here $vv^* f = v \langle v, f \rangle$. The sum includes every active prime power. All sums in R_L are finite. In the additive normalization of the Weil functional in [1, Section 1.1], Q_L is exactly the Weil form of $E_L f$. The constant, both poles and all prime powers are essential.

The digamma expansion [3] gives $b(\tau^2) = \log |\tau| - \log 2 - \psi(1/4) + O(\tau^{-2})$. Consequently $\mathcal{D}_L$ is the form domain and $C_c^\infty(0, L)$ is a form core. The support contraction followed by mollification proves the core assertion in this logarithmic Fourier norm. Bounded perturbation then defines the self-adjoint, semibounded W_L on $\operatorname{Dom} T_L$.

Let H_N and H_D denote the Neumann and Dirichlet interval Laplacians. Minimizing

$$\frac{2}{a} \left(\|E_L f - u\|_{L^2(\mathbb{R})}^2 + a^{-2} \|u'\|_{L^2(\mathbb{R})}^2 \right)$$

over $u \in H^1(\mathbb{R})$ gives $(2/a)s/(a^2 + s)$ in Fourier space. Restricting u to the interval lowers this minimum; allowing only zero extensions of $H_0^1(0, L)$ competitors raises it. Summation proves the closed-form bracket

$$b(H_N) \leq T_L \leq b(H_D). \quad (2.5)$$

The positive resolvent representation is a concrete case of the extension framework in [4]. The comparison does not assign Neumann boundary conditions to the actual whole-line fields.

Write

$$e_j(x) = \nu_j \cos(j\pi x/L), \quad \nu_0 = L^{-1/2}, \quad \nu_j = \sqrt{2/L} \ (j > 0), \quad b_j = b((j\pi/L)^2).$$

For $q \in (0, 1)$ define

$$r_h(q) = \max_{1 \leq i \leq h} \left(\sum_{j=1}^{i-1} q^j + \sum_{j=1}^{h-i} q^j \right),$$

and specify

$$\beta_L = -w_0 + 2 \sinh(L/2) - L + \sum_{\log p < L} (\log p) r_{\lceil L/\log p \rceil}(p^{-1/2}). \quad (2.6)$$

The prime bound follows by decomposing translation by $\log p$ into finite chains and bounding each symmetric correlation matrix by its largest row sum. The odd pole costs at most $2\|s_L\|^2 = 2\sinh(L/2) - L$; the even pole is nonnegative. Thus

$$W_L \geq b(H_N) - \beta_L I. \quad (2.7)$$

Fix $\delta_0 = 1/16$ once and for all and define

$$N(L) = \min\{N \geq 1 : b_N \geq \beta_L + \delta_0\}, \quad P = \sum_{j=0}^{N(L)-1} e_j e_j^*, \quad Q = I - P. \quad (2.8)$$

The set is nonempty because $b_j \rightarrow \infty$. Any larger certified cutoff gives an equivalent criterion. The high compression, defined by restricting the closed form to $Q\mathcal{D}_L$, obeys $H := W_L|_{Q\mathcal{D}_L} \geq \delta_0 I$. This bound is unconditional and makes H^{-1} a known bounded response. It says nothing yet about the low/high coupling.

3 An explicit boundary operator

The following representation makes that coupling accessible. For $a > 0$ set

$$u_a(x) = e^{-ax}, \quad v_a(x) = e^{-a(L-x)}, \quad r_a = e^{-aL}, \quad K_a = \frac{u_a u_a^* + v_a v_a^* + r_a(u_a v_a^* + v_a u_a^*)}{1 - r_a^2}.$$

Theorem 3.1 (Boundary representation). *The positive finite sums $K^{(J)} = \sum_{k < J} K_{a_k}$ converge strongly to a bounded positive operator K_L satisfying*

$$T_L = b(H_N) + K_L, \quad \text{Dom } T_L = \text{Dom } b(H_N), \quad \mathcal{D}_L = \text{Dom } b(H_N)^{1/2}. \quad (3.1)$$

Finite cosine sums are an operator core. The essential norm is $\|K_L\|_{\text{ess}} = \pi/2$, so finite mass sums do not converge in operator norm. Nevertheless, for the first h cosine modes,

$$\|(K_L - K^{(J)})P_h\| \leq \eta_{h,J} := \sqrt{\frac{2(2h-1)}{L}}(a_J^{-1/2} + a_J^{-3/2}) \rightarrow 0. \quad (3.2)$$

Proof. The compressed free Green function is $e^{-a|x-y|}/(2a)$; the Neumann Green function is

$$\frac{\cosh(a \min(x, y)) \cosh(a(L - \max(x, y)))}{a \sinh(aL)}.$$

Their difference multiplied by $2a$ is K_a . Subtracting the corresponding resolvent energies proves the finite-tower version of (3.1). The coefficient matrix of K_a is positive.

Let $n(t) = e^{-t/2}/(1 - e^{-2t})$. Summing the kernels gives $n(x+y) + n(2L-x-y)$ plus a bounded kernel. Since $n(t) = (2t)^{-1} + O_L(1)$ on $(0, 2L]$, this is one half of each endpoint Carleman kernel, modulo a bounded-kernel, hence compact, operator. The weighted Schur test bounds the Carleman operator by π . The partial towers have the same uniform kernel bound, so monotone convergence yields a bounded strong limit and the asserted domains.

Splitting the interval in half makes the two endpoint operators a direct sum modulo compact operators. Each has essential norm π before the factor $1/2$: normalized functions proportional to $t^{-1/2}$ on $[1, R]$, then dilated toward zero, have limiting Rayleigh quotient π and tend weakly to zero. This also proves the lower essential-norm bound; see the Carleman background in [5].

Direct integration, with $\omega_j = j\pi/L$, gives

$$K_a e_j = \frac{\nu_j a}{a^2 + \omega_j^2} (u_a + (-1)^j v_a). \quad (3.3)$$

Its norm is at most $\sqrt{2}\nu_j a^{-3/2}$. Sum the decreasing series using $\sum_{k \geq J} a_k^{-3/2} \leq a_J^{-3/2} + a_J^{-1/2}$, then sum squared column bounds over $j < h$ to obtain (3.2). $\square$

In particular all low basis vectors lie in $\text{Dom } W_L$. The boundary entries vanish for opposite parities; for equal parities they are

$$\langle e_i, K_L e_j \rangle = 2\nu_i \nu_j \sum_{k \geq 0} \frac{a_k^2 (1 - (-1)^j e^{-a_k L})}{(a_k^2 + \omega_i^2)(a_k^2 + \omega_j^2)}. \quad (3.4)$$

The remaining entries are elementary integrals of the rank-two pole kernel and of translated cosines on their overlap intervals. These formulas specify the operator using gamma and prime data, independently of RH.

4 Exact elimination and the equivalence theorem

For each L , use the prescribed P, Q and define

$$A = PW_L P, \quad B = QW_L P, \quad H = W_L|_{QX_L}.$$

Here A is an $N(L)$ by $N(L)$ Hermitian matrix, $B : PX_L \rightarrow QX_L$ is bounded, and H is the positive operator already bounded below in (2.7). Define the finite matrix

$$S_L = A - B^* H^{-1} B, \quad S_n := S_L|_{L=n}. \quad (4.1)$$

Its entries involve an infinite-dimensional inverse. It is a finite matrix with explicitly specified, approximable entries, not a formula involving only finitely many elementary operations.

Theorem 4.1 (Exact RH equivalence). *For each $L > 0$, $Q_L \geq 0$ on $\mathcal{D}_L$ if and only if $S_L \succeq 0$. Consequently Conjecture 1.1 is equivalent to RH.*

Proof. Write $f = p + h$, $p = Pf$, $h = Qf$. Completing the high square gives the identity of closed forms

$$Q_L[p + h] = \|H^{1/2}(h + H^{-1}Bp)\|^2 + \langle p, S_L p \rangle. \quad (4.2)$$

Every term is defined: $p \in \text{Dom } W_L$, $h \in \text{Dom } H^{1/2}$, and $H^{-1}Bp \in \text{Dom } H$. If $S_L \succeq 0$, this proves positivity. Conversely, choose $h = -H^{-1}Bp$ to recover exactly $\langle p, S_L p \rangle$.

RH implies Weil positivity on smooth compactly supported functions. The form core extends it to $\mathcal{D}_L$, so every $S_n \succeq 0$. For the converse, every compact support can be translated into $(0, n)$ for some integer n . The Weil form is translation invariant: its gamma multiplier and prime correlations are invariant, and the pole kernel is $2 \cosh((x - y)/2)$. Zero extension to a larger interval preserves all these terms. Thus positivity of all S_n implies Weil positivity on all required test functions, and hence RH by the classical criterion [1]. $\square$

No uniform gap in n , common source on all intervals, or compatible limit of the matrices is required. The dimensions can grow without bound.

4.1 A comparison between two positive matrices

To retain the positive-source language of the investigation, put $\alpha = \beta_L + 1$. Equation (2.7) proves $W_L + \alpha I \geq I$ without RH. Define

$$G_L = A + \alpha I - B^*(H + \alpha I)^{-1}B, \quad (4.3)$$

$$D_L = \alpha I + B^*(H^{-1} - (H + \alpha I)^{-1})B. \quad (4.4)$$

Minimizing the positive shifted form gives $G_L \geq I$, and resolvent monotonicity gives $D_L \geq \alpha I > 0$. Their difference is exactly S_L . Thus Conjecture 1.1 can equivalently be written

$$\lambda_{\max}(G_n^{-1/2}D_nG_n^{-1/2}) \leq 1 \quad (n \geq 1). \quad (4.5)$$

The positivity of G_L and D_L separately supplies no ordering between them. Changing an admissible scalar shift changes the two matrices but cancels from S_L .

5 Bounds on the unresolved matrix

Put $d_j = b_j - \beta_L$ for $j \geq N$, and let $\Lambda e_j = d_j e_j$ on QX_L . Then $H \geq \Lambda \geq \delta_0 I$. For any finite-input trial map $Y : PX_L \rightarrow \text{Dom } H$, set $R_Y = B - HY$. Expansion gives

$$B^*H^{-1}B = B^*Y + Y^*B - Y^*HY + R_Y^*H^{-1}R_Y, \quad (5.1)$$

$$0 \leq R_Y^*H^{-1}R_Y \leq R_Y^*\Lambda^{-1}R_Y. \quad (5.2)$$

For $K > N$, let $P_{N,K}$ project onto $e_N, \dots, e_{K-1}$ and define the Galerkin trial by the positive finite system

$$(P_{N,K}HP_{N,K})Y_K = P_{N,K}B, \quad \text{ran } Y_K \subset P_{N,K}X_L.$$

Its residual $R_K = B - HY_K$ is supported in cosine modes $j \geq K$.

Theorem 5.1 (Convergent enclosures). *The finite Hermitian matrices*

$$L_{L,K} = A - B^*Y_K - d_K^{-1}R_K^*R_K, \quad U_{L,K} = A - B^*Y_K \quad (5.3)$$

satisfy $L_{L,K} \preceq S_L \preceq U_{L,K}$ and converge to S_L in norm as $K \rightarrow \infty$. Their gap is exactly $d_K^{-1}R_K^*R_K$.

Proof. Galerkin orthogonality gives $Y_K^*HY_K = B^*Y_K = Y_K^*B$ and $P_{N,K}R_K = 0$. Apply (5.1) and $\Lambda^{-1} \leq d_K^{-1}I$ on the remaining modes to obtain the bounds. For convergence, write $H = \Lambda + E$ with E bounded using Theorem 3.1. The cosine core and coercivity imply $Y_K \rightarrow H^{-1}B$ in the form norm. If P_K contains the first K modes,

$$R_K = (I - P_K)(B - EY_K) \rightarrow 0$$

in operator norm on the finite input space. Therefore the enclosure gap tends to zero, proving the assertion. $\square$

The column tail (3.2) controls finite mass approximations to B and HY_K . The residual Gram must include their interference with the pole and prime columns. Positive eigenvalues of an upper enclosure alone are not a positivity certificate.

Corollary 5.2 (An equivalent tolerance formulation). *Conjecture 1.1, and hence RH, is equivalent to*

$$\forall n, m \in \mathbb{N} \exists K > N(n) : \quad L_{n,K} + 2^{-m}I \succ 0. \quad (5.4)$$

Proof. If $S_n \succeq 0$, choose K with $\|L_{n,K} - S_n\| < 2^{-m}$, using Theorem 5.1. Conversely $S_n \succeq L_{n,K} \succ -2^{-m}I$ for every m implies $S_n \succeq 0$. This avoids assuming that a semidefinite limit is detected by an unshifted, finite strict certificate. $\square$

The strict margin makes each true instance amenable to sufficiently accurate enclosures. The companion derivations give a version using arbitrary finite trials and explicit column-error allowances. No finite collection of successful intervals or tolerances establishes (5.4).

6 What makes the remaining conjecture arithmetic

The reduction works for many semibounded operators. For example $W = \begin{pmatrix} 1/2 & 1 \\ 1 & 1 \end{pmatrix}$ has a positive high block $H = 1$ and $W + I > 0$, but its low Schur complement is $-1/2$. A positive shifted source and a positive high sector alone therefore do not imply Conjecture 1.1.

There is a more relevant negative control. Define the Chebyshev function $\psi_{\text{Ch}}(x) = \sum_{p^m \leq x} \log p$ and $E(r) = \psi_{\text{Ch}}(e^r) - e^r + 1$. For $h_f(r) = \text{Re}\langle E_L f, U_r E_L f \rangle$, cancellation of the smooth prime density against the poles and the lowest gamma mass gives

$$Q_L[f] = t_{\geq 1, L}[f] + (4 + w_0)\|f\|^2 - 2 \int_0^L e^{-r/2} h_f(r) dE(r). \quad (6.1)$$

The Stieltjes form holds on $\mathcal{D}_L$; derivatives of h_f are not required. If the prime measure is replaced by $e^r dr$, the last term vanishes. On the unit input $f(x) = \sin(\pi x/2)$ on $(0, 2)$,

$$Q_2^{\text{dens}}[f] < -1 + \frac{26}{125} \frac{(22/7)^2}{4} = -\frac{2979}{6125} < 0.$$

Here $w_0 < -5$ and $\sum_{k \geq 1} 2/a_k^3 \leq 26/125$; smooth approximation preserves negativity. The actual prime discrepancy carries information that the leading density and the positive response architecture cannot replace.

At $L = 1$, the longer manuscript gives a rational certificate allowing $N = 4$ and $\delta_0 = 1/16$. That is a certificate for the high sector; it does not evaluate or establish the sign of S_1 . Present bounds may require very large dimensions as L increases. No practical complexity estimate, new positivity interval, or zero computation is asserted here.

The proposed short paper consists of the unconditional boundary and response theorems, the exact equivalence, and the open matrix conjecture. The extended supersymmetric models motivate the investigation but are not hypotheses of this equivalence. In particular, no unproved physical tt^* identification is being used to infer an arithmetic sign.

Preparation and verification status

This draft was prepared with substantial assistance from OpenAI's Codex in mathematical analysis, derivation checking, code and exposition. It is a draft for human mathematical review; completion of that review is not claimed. The companion file expands the analytic arguments and identifies the checks needed to support the equivalence. The repository citation records the source revision used here. A version-specific archival DOI has not yet been supplied.

References

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- [6] *Arithmetic ground-state geometry: Physical caps, arithmetic sources and analytical sign tests*, working manuscript in the [shifted-zeta-positivity](#) project, September 2026, [repository revision efb908a](#), especially Sections 10, 14 and 15.